cereals and millets(65-85%) and roots and tubers(19-35%). Starch is a polysaccharide formed in nature by the condensation of large number (4000-15000) of glucose molecules. It consists of a mixture of 2 components called amylose and amylopectin. It is the storage form of carbohydrate in the plant kingdom. Cooking facilitates the digestion of starch. Boiling causes swelling of the starch granules and rupture of the cell walls, allowing better digestion. The enzyme amylase present in the salivary and pancreatic juices, converts starch into maltose which is subsequently broken into glucose and absorbed.

2) **Dextrin :** It is not found in direct form in nature. They are polysaccharides formed by the partial hydrolysis of starch by acids or amylase. They are composed of large number of glucose molecules.

3) **Glycogen:** It is made up of chains of glucose,which are more highly branched than starch molecules.It is the storage form of carbohydrates in human beings and animals. It is formed by the condensation of large number (5000-10000) of glucose molecules.When required by the body, glycogen is converted to glucose to give energy.

ACTIVITY - 1

Match the right answer

a) Fruit sugar	-	Lactose
b) Grape sugar	-	Maltose
c) Milk	-	Starch
d) Malt sugar		Storage form of carbohydrate
e) Dextrin	-	Sucrose
f) Glycogen	-	Fructose
g) Cane sugar	-	Glucose

9.3 Functions of carbohydrates

The functions of carbohydrates in the body are as follows:

1) **Energy**: The principle function of carbohydrates is to serve as a major source of energy for the body. Each gram of carbohydrate yields 4kcal of energy regardless of its source.They provide an economical and quick source of energy. Excess carbohydrates in the body is stored as glycogen and can be converted to glucose for energy production when required .

2) **Protein Sparing Action:** Carbohydrates exert a protein sparing action. If sufficient amounts of carbohydrates are not available in the diet, the body will convert protein to glucose in order to supply energy. Hence, in order to spare proteins for tissue building and repair, carbohydrates must be supplied in optimum amounts in the diet. This is called the protein sparing action of carbohydrates.

3) **Fat Metabolism**: Adequate supply of carbohydrates determines the amount of fat to be metabolized for energy, which in turn affects the formation and disposal rate of ketones (intermediate products in fat metabolism). In the absence of adequate supply of carbohydrates, more fat is used because of which ketones accumulate in the body and this results in a disorder called Ketosis or Acidosis. This shows that carbohydrates have an anti-ketogenic effect which prevents harmful

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